

SEUNGPIIL LEE

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Personal Statement

Master’s researcher who moves in both directions between AI and the human mind: from AI back to the mind it imitates, and from the mind forward into the AI it shapes. My current focus is the cognition of intelligent systems, studied through LLMs — mechanistic interpretability keeps me honest about mechanisms, and memory-augmented agents that learn at test time are the current testbed. Coming in through literature and philosophy of mind, I draw on cognitive science and STS alongside the engineering.

EDUCATION

Gwangju Institute of Science and Technology (GIST) Master’s Program, Major in AI Convergence	<i>March 2025 – present</i>
Gwangju Institute of Science and Technology (GIST) B.S. in Electrical Engineering and Computer Science Minor in Mathematics, Minor in Literature	<i>March 2018 – December 2024</i> GPA: 3.66/4.50
UC Berkeley Berkeley Summer Session Program	<i>June 2019 – August 2019</i> GPA: 4.00/4.00

PUBLICATIONS

Journal Articles

- **Seungpil Lee***, Woochang Sim*, Donghyeon Shin*, Sanha Hwang, Wongyu Seo, Jiwon Park, Seokki Lee, Sejin Kim, and Sundong Kim. “Reasoning Abilities of Large Language Models: In-Depth Analysis on the Abstraction and Reasoning Corpus.” *ACM Transactions on Intelligent Systems and Technology (ACM TIST)*, 2024.
- Sanha Hwang, **Seungpil Lee**, Sejin Kim, and Sundong Kim. “Solution Augmentation for ARC-AGI Problems Using GFlowNet: A Probabilistic Exploration Approach.” *Transactions on Machine Learning Research (TMLR)*, 2025.

Conference Papers

- Heejun Kim*, **Seungpil Lee***, Jewon Yeom, Jaewon Sok, Seonghyeon Park, Jeongjae Park, Taesup Kim, and Sundong Kim. “From Noise to Diversity: Random Embedding Injection in LLM Reasoning.” *Conference on Neural Information Processing Systems (NeurIPS)*, 2026. (Earlier version at ICML Mechanistic Interpretability Workshop 2026, Spotlight)
- **Seungpil Lee**, Donghyeon Shin, Yunjeong Lee, and Sundong Kim. “Can Large Language Models Develop Gambling Addiction?” *Conference on Neural Information Processing Systems (NeurIPS)*, 2026.
- **Seungpil Lee***, Donghyeon Shin*, Klea Lena Kovacec, and Sundong Kim. “From Generation to Selection: Findings of Converting Analogical Problem-Solving into Multiple-Choice Questions.” *Findings of EMNLP (EMNLP Findings)*, 2024.
- Hosung Lee*, Sejin Kim*, **Seungpil Lee**, Sanha Hwang, Jihwan Lee, Byung-Jun Lee, and Sundong Kim. “ARCLE: The Abstract and Reasoning Corpus Learning Environment for Reinforcement Learning.” *Conference on Lifelong Learning Agents (CoLLAs)*, 2024.

* indicates equal contribution

SKILLS

Programming	C, C++, Java, JavaScript, Python
Tools	LaTeX, Spring Boot, PyTorch, Hugging Face
Language	Korean (Native), English (Intermediate)

EXPERIENCE

LG AI Research, Seoul

September 2026 – present

Research Intern

- Improving the coding capability of K-Exaone, a Korean sovereign foundation model

Microsoft Research Asia (MSRA), Shanghai

February 2026 – August 2026

Research Intern, Collaborative Research Program

- Developed test-time skill optimization for LLM agents, letting them acquire and reuse task skills at inference time without parameter retraining
- Extended the Agent-Lightning framework with memory generation so agents carry learned context across long-horizon tasks
- Bridged mechanistic analysis of reasoning failures with memory-based mitigations

DataScience Lab, GIST

September 2023 – present

Undergraduate Intern → Master's Student

- LLM skill discovery and evaluation with cognitive science tools (Language of Thought Hypothesis)
- LLM alignment using RLVR methods; Mechanistic interpretability with Sparse Autoencoders

Teaching Assistant, Computer Networking

September 2023 – December 2025

- Designed assignments connecting theory to implementation; SDN demo with Ryu and Mininet

Development Team, Republic of Korea Navy

January 2022 – September 2023

- Mandatory military service; back-end developer for internal app and web services

BioComputing Lab, GIST

June 2020 – January 2021

Undergraduate Intern

- Developed Spiking Neural Network methods with synthetic gradients

AWARDS & FUNDING

National Research Foundation of Korea (NRF) Funding

Sep 2025 – Sep 2026

Research funding for outstanding master's students

Best Undergraduate Thesis Award, GIST EECS

February 2025

Awarded for graduation thesis on MC-LARC benchmark

Korean Government Scholarship, GIST College

March 2018 – August 2024

Full scholarship for students at GIST

Scholarship for Summer Session Abroad

June – August 2019

UC Berkeley Summer Session scholarship

Navy AI Competition

August 2023

Award in drone object detection competition

K-StartUp Finals

November 2023

Advanced to finals as ML researcher and back-end developer